A complete thirteen-variable consistency equation

Strictly positive integer unknowns; exact degree 51; a 90-operation evaluator.

The only unknowns are

$$j_*, F, \alpha, f, i, T, X, Y, r, \eta, \zeta, u, \gamma \in \mathbb{Z}_{>0}.$$

All other symbols in the equation are polynomial abbreviations:

$$\begin{gathered}
J = 3j_* + 1, \quad \bar{q} = \beta J, \quad q = \bar{q} + 1, \quad C = q - F - \alpha, \\
E = XY, \quad k = \eta + \zeta, \quad c = kY + \eta, \quad a = E + Y, \\
H = 4a + 3, \quad \Delta = a^2 + H, \quad U = q^2 Y, \\
M = q^3 \bar{q}, \quad \ell = E(kY), \quad W = \bar{q}k + 1, \\
B_t = (qK + X)C - qF, \\
R = (\bar{q}(q - F) + \alpha)(q^2 - 1) + (M_C + qM_F)J, \\
Z = M(\ell + k + r) + \bar{q}Y - q^3 W B_t, \\
D = U(X + ac) + H\{U\gamma + qE - (k - R - 1)\}, \\
A_0 = Z^2 - M^2\{\ell(\ell + k) + 1\}, \\
A_1 = D^2 - U^2(\Delta c^2 + 1), \\
Q = \Delta(f^2 - 1), \quad I = ic^2, \\
V = f^2(cT - R) - c, \quad y = V + u, \\
F_a = y^2 - Q(y^2 - V^2), \quad F_s = 1 + (I^2 - Q).
\end{gathered}$$

The single equation is

$$\boxed{(1 + A_0)(1 + A_1)F_a F_s - 1 = 0.}$$

The four fixed integers. Every index, set, sum, product and floor below defines a constant, not another unknown. Put

$$\rho = 2^{67}, \quad B = \rho^{8807}, \quad \beta = B - 1, \quad G_m(x) = \sum\nolimits_{j=0}^{m-1} x^j.$$

Define

$$\begin{aligned}
\mathcal{E} = {} & \{0, \dots, 942\} \cup \{983, \dots, 992\} \cup \{1003, \dots, 1011\} \\
& \cup \{1013, \dots, 1031\} \cup \{1063, 1064, 1095, 2195, 3296\}.
\end{aligned}$$

Then

$$\begin{aligned}
M_C = {} & B - 1 - \sum\nolimits_{e \in \mathcal{E} \setminus \{1\}} \rho^e - 2\rho^{1063}, \\
M_F = {} & B - 1 + 4 + (2^{51} - 2)(\rho^{4393} + \rho^{5495}) \\
& + \rho^{1013} G_{10}(\rho) + \rho^{1065} G_9(\rho) + \rho^{2195} + \rho^{3296} \\
& + (2^{65} - 1)\rho^{8792} G_{13}(\rho) + (2^{19} - 1)\rho^{8805}, \\
K = {} & \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2201} + \rho^{8849} \\
& + (\rho^{4393} + \rho^{5495})\Phi(1/\rho),
\end{aligned}$$

where

$$\begin{aligned}
\Phi(x) = {} & (1 + 2^{51}) G_{943}(x) - 2^{10} \sum\nolimits_{t \in \mathcal{T}} v(t) x^{\varrho(t)} \\
& + \sum\nolimits_{(b,e,m) \in \mathcal{C}} 2^b x^e G_m(2x), \\
\mathcal{C} = {} & \{(10, 1023, 8), (18, 1013, 9), (27, 1003, 8), \\
& (35, 983, 9), (44, 1031, 1), (45, 1022, 1), (46, 1011, 1), \\
& (47, 992, 1), (48, 1095, 1), (49, 2195, 1), (50, 3296, 1)\}.
\end{aligned}$$

The rational intermediate in K cancels exactly; all four results are positive integers.

The 943 fixed tiles and their ranks. Use the actions u_j, v_j, d_j decoded below. Let

$$\begin{gathered}
h(q, b) = 2(q + 1) + b, \quad \mathcal{R} = \{u_j : d_j = 1,\ 0 \le j < 430\}, \\
\mathcal{L} = \{u_j : d_j = 0,\ 0 \le j < 430\}, \\
r_q = 6 + |\mathcal{R} \cap \{0, \dots, q - 1\}|, \quad l_q = 205 + |\mathcal{L} \cap \{0, \dots, q - 1\}|.
\end{gathered}$$

An interval ending at -1 is empty. The following union is $\mathcal{T}$; the index ranges in each row are fixed:

range	tile (w, n, e, s)
$j = 2q + b < 430$	$((1 - d_j)l_{u_j}, h(q, b), d_j r_{u_j}, v_j)$
$b = 0, 1$	$(0, h(215, b), 0, h(215, b)),\ (0, b, 0, b)$
$q \in \mathcal{R},\ b = 0, 1$	$(r_q, b, 0, h(q, b))$
$q \in \mathcal{L},\ b = 0, 1$	$(0, b, l_q, h(q, b))$
$t \in \{0, 1, 432, 433\}$	$(0, t, 0, 438)$
	$(2, 434, 2, 0),\ (2, 435, 3, 1)$
	$(3, 436, 4, h(u_0, 0)),\ (4, 436, 4, 0),\ (5, 0, 0, 0)$
$t = 434, 435, 436$	$(1, 438, 1, t)$
$a \in \{0, 1, 4\},\ b = 0, 1, 2$	$(a, 437, b, 437)$

Put $s_* = (2, 435, 3, 1)$, $f_* = (5, 0, 0, 0)$, and $\mathcal{T}' = \mathcal{T} \setminus \{s_*, f_*\}$. For $t = (w, n, e, s)$ set

$$\begin{gathered}
\kappa(t) = 2^{30} w + 2^{20} n + 2^{10} e + s, \\
v(t) = w + 2^8 n + 2^{17} e + 2^{25} s + 127\, 2^{34}, \\
\varrho(s_*) = 0, \quad \varrho(f_*) = 1, \\
\varrho(t) = 2 + \sum\nolimits_{z \in \mathcal{T}'} \left\lfloor \frac{2^{40} + \kappa(t) - \kappa(z) - 1}{2^{40}} \right\rfloor \quad (t \in \mathcal{T}').
\end{gathered}$$

A closed decoder for the 430 actions. Put $D_b(A, k) = \lfloor A/b^k \rfloor \bmod b$, where $a \bmod b = a - b\lfloor a/b \rfloor$. For $0 \le q < 215$, $b = 0, 1$, $j = 2q + b$:

$$\begin{gathered}
N = \mathcal{A} \bmod 2^{636}, \quad T_0 = \lfloor \mathcal{A}/2^{636} \rfloor \bmod 2^{1675}, \\
E_0 = \lfloor \mathcal{A}/2^{2311} \rfloor, \\
\nu_j = D_2(N, j), \quad \sigma_j = \sum\nolimits_{k=0}^{j-1} \nu_k, \\
m_q = D_2(N, 430 + q), \quad \tau_q = \sum\nolimits_{k=0}^{q-1} m_k, \\
F_q = 14(1 - m_q) + m_q D_{16}(E_0, \tau_q), \\
u_j = \nu_j(1 + \sigma_j) + (1 - \nu_j) D_{216}(T_0, j - \sigma_j), \\
v_j = D_2(F_q, 2b), \quad d_j = D_2(F_q, 2b + 1).
\end{gathered}$$

Empty sums are zero. Here $(v_0, d_0, u_0) = (1, 1, 1)$; the constructor tile uses this actual first target, including its state renaming.

The integer $\mathcal{A}$. The alphabet, in order of digit values $0, \dots, 61$, is `0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz`.

Concatenate the following 436 characters as digits $e_0, \dots, e_{435}$; put $\mathcal{A} = \sum_{j=0}^{435} e_j 62^{435-j}$.

```
BxEnhvy6NipzkWmROo0GsA3uK3twWLxscLq2OUTumy5Cdu461LwD6g7RDNzF8Y
iYDFTpB8ehAypd7d6OsOL4APamnJeNXQypUezCmW9n4oXP6XfgmiWuAWZ8ZJ7U
OP7gu4kVdlGZqZHULQMar1dXNeJW4i7oNVI8usp8BlovjjZbwPCovZI328fRUq
4q92R81s5GjWQaQfGhkq9UYMK3rRMaBmsOIBkJqc2cj2VRIZ6mjzkBIOkW9SjZ
FzeA6J5u4j3sBL3SdgwXzM9mHvQivu8EJ1TBTpRhHDqZIbmYgwQ8VqTAFotl6F
We1YC554GKrVHsA8oUGKSA6F6k5KREBSDOTgd7bFALY84LLbpw574BQyvdoHrl
0Oe35uwDf15DpT6VJWZBjSbHPpPAINYTte0eUxnTFTKBlGj8Am6sb1gLldh8WZ
8P
```

The ordinary state/color-conjugacy, native and Pell arguments give unsolvability iff Con(ZFC). Native calibration and internal-ZFC formalization remain open in Lean. This page is a complete finite specification, not an unsolvability proof.